SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 **Total Marks: 25**

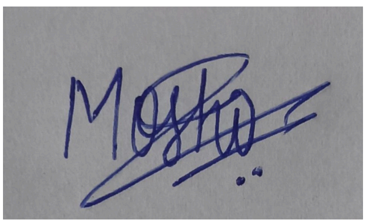
Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1. Practical significance, dataset, and literature: review: The chosen topic holds strong practical importance as it addresses real-world problems such as disease prediction, disaster management, or intelligent automation. The datasets considered should be relevant, diverse, and sufficiently large to ensure reliable model performance. A comprehensive literature review must include recent research works, methodologies, and comparative analysis, highlighting existing gaps and justifying the need for the proposed approach. This ensures the work is grounded in prior knowledge while contributing meaningful advancements. (5 Marks)

2. Implementation of preprocessing, post-processing, and evaluation: Effective implementation begins with data preprocessing steps such as data cleaning, handling missing values, normalization, and feature selection to improve model efficiency. Post-processing techniques, including result refinement, threshold tuning, and visualization, help in better interpretation of outputs. Evaluation procedures must include appropriate metrics like accuracy, precision, recall, and F1-score, along with validation techniques such as cross-validation to ensure robustness and generalizability of the developed model. (5 Marks)

3. Performance analysis and merit of the system: The performance of the developed system should be demonstrated using quantitative metrics and comparative analysis with existing methods. High accuracy, improved sensitivity, and reduced error rates indicate the effectiveness of the model. Visualization tools such as confusion matrices, ROC curves, and performance graphs can further support the findings. The results should clearly highlight how the proposed system outperforms traditional or baseline approaches in terms of efficiency, reliability, and scalability.
(5 Marks)

4. Future scope and publication potential: The future scope of the work may include extending the model to larger datasets, integrating real-time data processing, or applying advanced techniques such as deep learning and hybrid models. Improvements in interpretability and deployment can also be explored. The presented work can be converted into a research paper by structuring it with a clear abstract, methodology, results, and discussion, followed by proper citations and formatting as per journal or conference guidelines. (5 Marks)

5. Learning Problems and Perspectives: Explain different types of learning problems in machine learning, including supervised, unsupervised, and reinforcement learning. Discuss various perspectives such as symbolic and statistical approaches. Highlight

key issues like data quality, overfitting, and computational complexity, and explain how these factors influence the design and performance of learning systems.

(5 Marks)

6. Concept Learning and Hypothesis Representation: Describe the concept learning task and its importance in machine learning. Explain how hypotheses are represented and evaluated. Discuss general-to-specific ordering and the role of training examples in refining hypotheses. Provide insights into how concept learning helps in classification problems and knowledge extraction. (5 Marks)

7. Version Spaces and Candidate Elimination: Explain the concept of version spaces and how they represent all hypotheses consistent with training data. Describe the Candidate Elimination algorithm, including the roles of specific (S) and general (G) boundaries. Discuss how these boundaries are updated with positive and negative examples during the learning process. (5 Marks)

8. Inductive Bias in Learning Algorithms: Define inductive bias and explain its significance in machine learning. Discuss different types of bias such as restriction bias and preference bias. Explain how inductive bias helps a learning algorithm generalize from limited data and influences hypothesis selection in concept learning.

(5 Marks)

9. Decision Tree Learning and Representation : Explain the working of decision tree learning algorithms and their representation of knowledge. Discuss concepts such as entropy, information gain, and tree pruning. Highlight advantages and limitations of decision trees, and explain how they are used for classification and decision-making tasks. (5 Marks)

10. Heuristic Search in Machine Learning: Describe heuristic search and its role in machine learning, especially in hypothesis space exploration. Explain how heuristics guide the search process to find optimal or near-optimal solutions efficiently. Discuss examples where heuristic search improves learning performance and reduces computational complexity. (5 Marks)

11. Explain the basic structure of **artificial neural networks (ANNs)**, including input, hidden, and output layers. Discuss activation functions, backpropagation, and gradient descent. Highlight the advantages of deep learning over traditional machine learning methods and its applications in real-world problems. (5 Marks)

12. Describe the working principle of **Support Vector Machines (SVMs)** for classification tasks. Explain the concept of hyperplanes, margin maximization, and support vectors. Discuss kernel functions and how they enable SVMs to handle non-linear data. (5 Marks)

13. Explain the concept of ensemble learning and its importance in improving model performance. Discuss techniques such as bagging, boosting, and stacking. Highlight algorithms like Random Forest and AdaBoost, and explain how combining multiple models reduces variance and bias. (5 Marks)

14. Describe the importance of feature selection in machine learning. Explain different techniques such as filter methods, wrapper methods, and embedded methods. Also, discuss dimensionality reduction techniques like Principal Component Analysis (PCA) and their role in improving model efficiency and performance.

(5 Marks)

15. Explain various model evaluation techniques such as train-test split, cross-validation, and bootstrapping. Discuss performance metrics for classification and regression problems. Highlight the importance of proper validation in avoiding overfitting and ensuring model generalization. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**35**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**25**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**10**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Neural Networks: The Foundation of Deep Learning

ITA0605 - MACHINE LEARNING

PRESENTED BY :

MOSHIKA M (192424064)

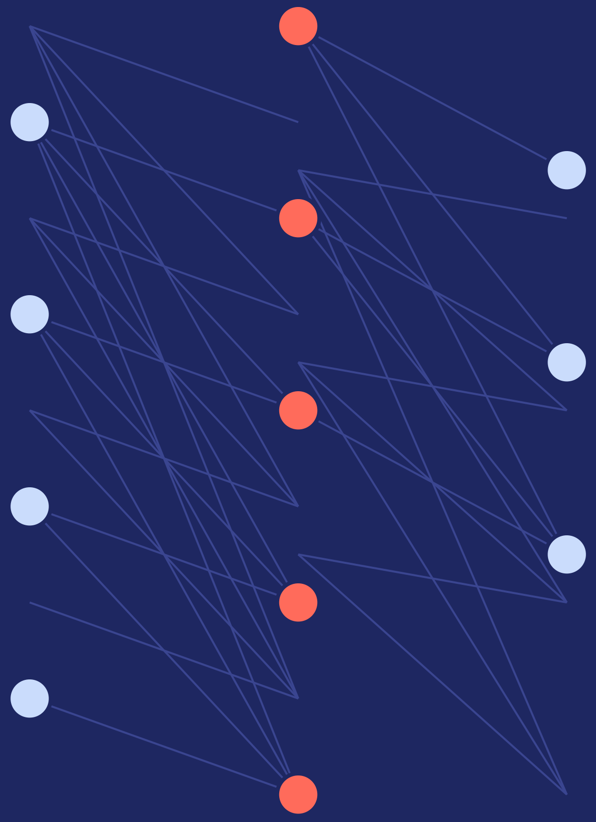
GUIDED BY:

Dr. KAVITHA

FROM NEURONS TO INTELLIGENCE

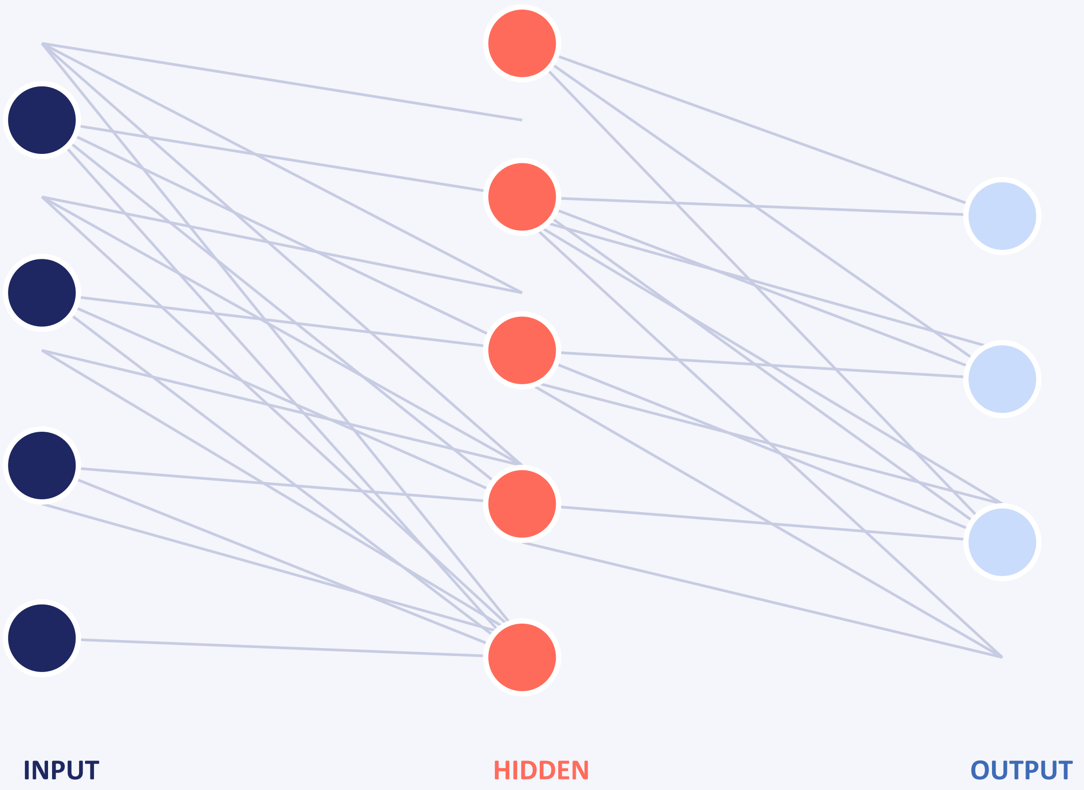
ARTIFICIAL NEURAL NETWORKS

How layered computation, activation functions, and gradient-based learning power modern deep learning systems



Inside the Network: Layer by Layer

Every ANN is built from three kinds of layers that pass signals forward



Input Layer

Receives raw data — pixels, words, or sensor readings — one node per feature.

Hidden Layer(s)

Transform inputs through weighted sums and activation functions, extracting increasingly abstract patterns.

Output Layer

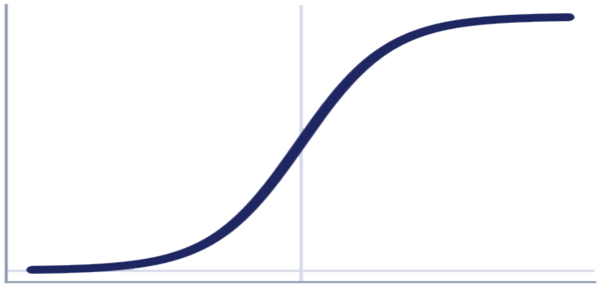
Produces the final prediction — a class label, probability, or numeric value.

Activation Functions: Adding Non-Linearity

Without them, a deep network collapses into a single linear function — activations let it model complex patterns

Sigmoid

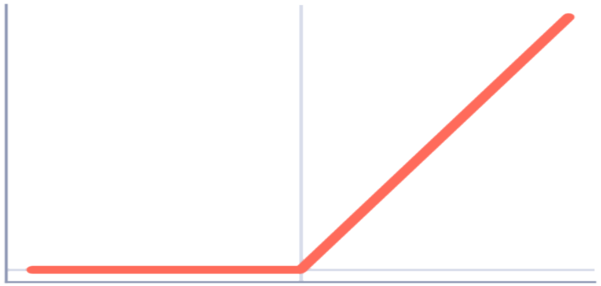
$\sigma(x) = 1 / (1 + e^{-x})$



Squashes values into (0, 1). Useful for probabilities, but saturates and slows learning on deep nets.

ReLU

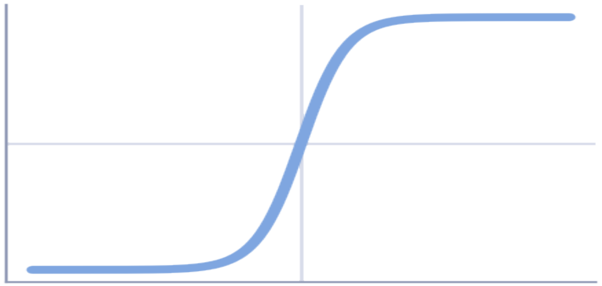
$f(x) = \max(0, x)$



Fast, sparse, and avoids vanishing gradients. The default choice for hidden layers today.

Tanh

$\tanh(x) = (e^{x}-e^{-x})/(e^{x}+e^{-x})$



Zero-centered output in (-1, 1); often trains faster than sigmoid in earlier hidden layers.

How Networks Learn: Backpropagation & Gradient Descent

Training is a repeated four-step loop that nudges every weight toward lower error

1

Forward Pass

Input flows through the network, layer by layer, to produce a prediction.

2

Compute Loss

A loss function measures how far the prediction is from the true value.

3

Backward Pass

The chain rule propagates the error gradient back through every layer.

4

Update Weights

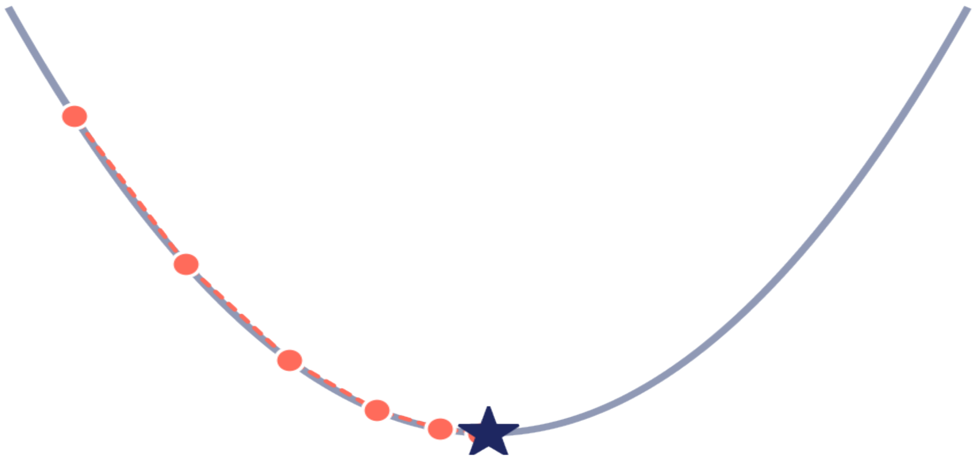
Gradient descent shifts each weight opposite its gradient, shrinking the loss.

Gradient Descent, Visually

The loss surface forms a bowl; the gradient points uphill, so each step moves **opposite** the gradient.

$$\theta \leftarrow \theta - \eta \cdot \nabla L(\theta)$$

Here η (the learning rate) controls step size: too small and training crawls; too large and it overshoots the minimum.



Why Deep Learning? Advantages & Real-World Impact

ADVANTAGES OVER TRADITIONAL ML

Automatic Feature Learning

Learns relevant features directly from raw data instead of relying on hand-engineered ones.

Scales With Data

Accuracy keeps improving as more data and compute are added, unlike classical models that plateau.

Handles Unstructured Data

Excels at images, audio, and text where traditional ML struggles without heavy preprocessing.

End-to-End Learning

Optimizes the entire pipeline jointly, from raw input to final decision, in one training process.

REAL-WORLD APPLICATIONS

Computer Vision

Image & face recognition

NLP

Translation, chatbots

Speech Recognition

Voice assistants

Autonomous Vehicles

Perception & control

Healthcare

Diagnostics, imaging

Recommenders

Retail, streaming

THANK YOU!